Week 49 Four towers and differences

Facilitator guide | Final student packets v2 | Prepared 3 October 2026 | Revised 23 October 2026 | Unpiloted

Mathematical overview

Simultaneous cyclic differences. With stations A,B,C,D in arrow order, the rule is D(a,b,c,d) = (|a-b|, |b-c|, |c-d|, |d-a|). All four outputs use the same old state. The gap on an arrow is assigned to its tail station, including D to A. Updating a tower in place before computing the others is a different process.

For nonnegative integer starts, **every four-station run reaches zero**. The maximum never increases. Four rounds make all entries even; scaling commutes with the rule, so after 4k rounds every entry is divisible by 2^k. Once 2^k exceeds the original maximum, boundedness forces every entry to be zero. The maximum need not decrease at each move, and the total height is not conserved.

Limits and contrast. Three stations can enter a nonzero cycle. The argument uses integer heights; do not claim that arbitrary real starts satisfy this termination proof. Scaling by a positive integer preserves the number of moves to zero. A zero start takes zero moves; a nonzero constant start takes one.

Discovery by band. K-1 finds gaps physically and searches for short or long runs. Grades 2-3 tests predecessor obstructions and effects of changing station order. Grades 4-5 relates the parity shadow, scaling and a bound to a universal termination proof. Many successful trials are evidence, not the general proof.

Prepare and launch

Per pair: 40 linking cubes, the student pages and pencils. Print at actual size. The paired working mats on pages 2 and 4 have 36 mm square stations, centers 52 mm apart along a side (three-station centers span 52 mm across and down), with a 9 mm gap and an arrow between the old and new mats. Small printed rings are reference diagrams, not cube-working spaces. Forty cubes hold an old and a new ring with heights up to 5, and the Grades 4-5 doubled start (4,10,2,0) with its next ring needs 36. The record rows keep the run: four boxes A-B-C-D, or A-B-C for three stations.

Readiness. K-1 matches cubes to find the unmatched part, with an adult reading and helping with the record rows. Grades 2-3 needs small absolute differences. Grades 4-5 needs even/odd and reasoning about repeated scaling; the final proof is optional for children who are still productively exploring.

Launch, 5 minutes. First let children compare two towers. Build the printed old ring (1,4,2,0); keep it untouched while making new A=3 from A and B. Complete new B=2, C=2 and D=1 together. Then clear the old mat and move each new tower to the same letter on it. Point to the wraparound arrow and keep labels fixed.

Flexible hour. 0-5 running around; 5-10 launch; 10-27 run starts on the mats; 27-47 choose longevity, predecessors or three stations; 47-55 share findings or parity explanations; 55-60 reset. Keep one adult anchored to each of the three tables.

Key for the opening runs

Tuples below always use the printed station order. Each arrow counts one move, including the move that first produces all zeros.

Problem 1, all bands.
(1,1,0,0) → (0,1,0,1) → (1,1,1,1) → zero: **3 moves**.
(1,0,1,0) → (1,1,1,1) → zero: **2 moves**.
(2,1,0,1) → (1,1,1,1) → zero: **2 moves**.
(0,0,0,2) → (0,0,2,2) → (0,2,0,2) → (2,2,2,2) → zero: **4 moves**, longest of the four printed starts.

Problem 2, all bands. The start (0,1,2,4) is allowed both for K-1's 0-4 range and the older 0-5 range, and takes **7 moves**:
(0,1,2,4) → (1,1,2,4) → (0,1,2,3) → (1,1,1,3) →
(0,0,2,2) → (0,2,0,2) → (2,2,2,2) → zero.
Independent exhaustive checks of all 625 and 1,296 permitted starts, respectively, give maximum 7. The prompt asks children to improve their attempts, not to reproduce that exhaustive proof.

Key for one and two moves

K-1 Problem 3. Exactly the positive constant starts (c,c,c,c) reach zero for the first time in one move. For example, (1,1,1,1) and (3,3,3,3). Every neighboring gap must be zero, so the whole ring must be constant. The page-1 rules say an all-zero start takes 0 moves, so (0,0,0,0) does not count.

K-1 Problem 4. Examples (0,1,0,1) and (0,1,2,1) both become (1,1,1,1), then zero. In general, a two-move start has all four neighboring gaps equal to the same positive c. Positivity makes the first output nonzero; being constant makes the next output zero.

Older Problem 3, in printed reading order.
(2,0,2,0): **possible**, from (0,2,2,0).
(1,1,1,0): **impossible**. Every output has an even total: modulo 2, each old height occurs in two gaps, so their sum is 0.
(1,1,1,1): **possible**, from (0,1,0,1).
(0,0,0,2): **impossible**. The first three zero gaps force a=b=c=d, making the fourth gap zero too. Even total is necessary, not sufficient.

Key for changing the number of stations

K-1 Problem 5; older Problem 4.
Four stations: (1,0,0,0) → (1,0,0,1) → (1,0,1,0) → (1,1,1,1) → zero.
Three stations: (1,0,0) → (1,0,1) → (1,1,0) → (0,1,1) → (1,0,1). The last three states form a cycle of period 3. The entire labeled state repeats, not merely its largest height or total. Determinism then forces this cycle to continue.

Key for station order and size

K-1 Problem 6; Grades 2-3 Problem 5. Both printed orders last 4 moves: (0,2,1,4) → (2,1,3,4) → (1,2,1,2) → (1,1,1,1) → zero, and (0,1,4,2) likewise. The order (0,1,2,4) lasts 7 moves (Problem 2's run). All rotations and reversals of (0,1,2,4) also last 7. They are the eight longest labeled orders; the other sixteen permutations last 4. To check completeness, rotate 0 to A and identify reversal pairs: only three representatives remain, (0,1,2,4), (0,1,4,2), (0,2,1,4), with durations 7,4,4.

Grades 2-3 Problem 6; Grades 4-5 Problem 7. A gap between two numbers in [0,M] is at most M. Applying this at every move shows that no later height exceeds the original maximum. Equality can persist: (0,1,2,4) becomes (1,1,2,4), with maximum 4 in both states. The page gives two blank rings for the child's own example. The rule does not say that each individual tower decreases.

Key for the parity and scaling investigations

Grades 4-5 Problem 5. Only **EEEE** is possible after four moves. Write E=0, O=1 and add modulo 2: |a-b| has the same parity as a+b. From any (a,b,c,d), the second parity row is (a+c,b+d,a+c,b+d), the third is (s,s,s,s) with s=a+b+c+d, and the fourth is (0,0,0,0). This symbolic argument covers all 16 starting parity patterns. The page has four columns of letter boxes, each a start and four moves; children can also write the 16 starting patterns on scraps and sort them.

Grades 4-5 Problem 6. The first printed start runs
(2,5,1,0) → (3,4,1,2) → (1,3,1,1) → (2,2,0,0) → (0,2,0,2) → (2,2,2,2) → zero.
The doubled start runs
(4,10,2,0) → (6,8,2,4) → (2,6,2,2) → (4,4,0,0) → (0,4,0,4) → (4,4,4,4) → zero.
Each state is doubled, because $|2a-2b|=2|a-b|$. Tripling works by the same identity with 3. Both runs take 6 moves.

Grades 4-5 Problem 8. Let M be the original maximum. Problem 5 makes $D^4(h)$ even. If $D^{4k}(h)=2^k v$, then four further moves give $2^k D^4(v)$, whose entries are divisible by 2^{k+1}. Hence $D^{4k}(h)$ is divisible by 2^k for every k. Problem 7 bounds each entry by M. Choose k with $2^k > M$: the only nonnegative multiple in [0,M] is zero. This proves finite termination. The bound 4k is convenient, not sharp.

Optional hints and sources

For an incorrect run, preserve both states and check one arrow at a time. For two-move starts, ask what the penultimate ring must look like. For parity, show one E/O conversion before asking for a classification. If the final proof stalls, ask separately what can grow and what divisibility accumulates; do not make the proof a prerequisite for enjoying the experiment.

Daniel Shapiro, *The Four-Number Game* (2005), pp. 1-3: author's explanation and parity proof. The simultaneous labeled-mat presentation, exact final trajectories, permutation classification and bounded searches were independently verified for this packet. The general proof is separate from finite enumeration. Preparation and pacing are unpiloted proposals, with no physical-test claim.

Starting routes and return visits

Planning update of 4 October 2026. The timing menus are examples; a theme can continue over several meetings. Keep the full Grades 4-5 investigations and choose by readiness.

Use and return choices

First routes: K-1 P1, then either P3-4 for one-/two-move starts or P2 for a longevity search; grades 2-3 and 4-5 P1-2. On return, K-1 can compare three stations or change the order (P5-6). Grades 2-3 can investigate predecessors (P3), station count (P4), or order and maximum bounds (P5-6). Grades 4-5 can make parity shadows (P5), then connect scaling and boundedness to termination (P6-8). Retain all upper proofs for deeper visits. Always compute four outputs from one untouched old state, then clear the old mat and move the new towers onto it.